Financial Intelligence using RAG-Enhanced LLMs

A Unified Explainable AI System for Indian Banking & Finance

1. PROBLEM STATEMENT

Large Language Models (LLMs) are increasingly applied to financial forecasting; however, they suffer from a critical flaw: hallucination — generating plausible but factually incorrect outputs. In high-stakes domains such as the Indian equity market. Existing approaches either rely solely on historical price signals (ignoring qualitative information) or use unconstrained LLMs (ignoring factual grounding). There is no widely adopted framework that grounds LLM reasoning in verified, retrieved financial evidence while also leveraging quantitative time-series signals.

2. OVERVIEW

This project proposes a hybrid AI forecasting system that combines Retrieval-Augmented Generation (RAG) with deep learning time-series models to deliver accurate, explainable, and stock price predictions for five major Indian equities. The system fuses three information streams:

- Quantitative signals — OHLCV price data with 20+ engineered technical indicators (SMA, EMA, RSI, MACD, Bollinger Bands, ATR).
- Sentiment signals — Polarity scores extracted from 50,000+ Indian financial news headlines using NLP (TextBlob / FinBERT-style pipeline).
- Contextual grounding — A FAISS-based RAG engine that retrieves top-k historically relevant news at inference time, constraining the LLM to factually grounded context and reducing hallucination.

The final forecast engine outputs: predicted next-day price, estimated volatility, risk label (Low / Medium / High), directional accuracy, and a natural-language contextual explanation.

3. MODELS & TECHNIQUES USED

Component	Model / Method	Purpose
Time-Series Model	LSTM + GRU (TensorFlow/Keras)	Sequential price & indicator forecasting
Sentiment Analysis	TextBlob / FinBERT	Financial news polarity scoring
Text Embedding	TF-IDF (512-dim)	News vectorisation for retrieval
Vector Database	FAISS (IndexFlatIP)	Cosine-similarity news retrieval
RAG Framework	Custom RAG Pipeline	Hallucination mitigation via grounding
Risk Scoring	Rule-based + ML	RSI, volatility & sentiment fusion
Scaling / Normalisation	MinMaxScaler (sklearn)	Feature range normalisation [0,1]

4. DATASETS

Stock Price Data — NSE-listed equities (Yahoo Finance), 2003–2025:

- **HDFC Bank (HDFCBANK.NS):** https://finance.yahoo.com/quote/HDFCBANK.NS/history/
- **ICICI Bank (ICICIBANK.NS):** https://finance.yahoo.com/quote/ICICIBANK.NS/history/
- **Infosys (INFY.NS):** https://finance.yahoo.com/quote/INFY.NS/history/
- **Reliance Industries (RELIANCE.NS):** https://finance.yahoo.com/quote/RELIANCE.NS/history/
- **Tata Consultancy Services (TCS.NS):** https://finance.yahoo.com/quote/TCS.NS/history/

News Data — Indian Financial News Headlines, 2003–2020 (Kaggle):

- **Indian Financial News Articles (2003–2020):** https://www.kaggle.com/datasets/hkapoor/indian-financial-news-articles-20032020?resource=download
  - 50,000+ headlines with publication date, title, and description
  - Covers major Indian corporates across banking, IT, and energy sectors

5. SYSTEM PIPELINE

Step	Stage	Key Actions
1	**Data Ingestion**	Load OHLCV CSVs + news CSV; handle missing values; forward-fill gaps; remove duplicates.
2	**Feature Engineering**	Compute SMA-10/20/50, EMA-12/26, MACD, RSI-14, Bollinger Bands, ATR, lag features (1/3/5/10 days), rolling mean/std, volume ratio, log returns.
3	**Sentiment Analysis**	Tokenise news text → TextBlob polarity scores → daily sentiment aggregation (mean, 7-day MA) → merge with stock data by date.
4	**RAG Index Construction**	TF-IDF vectorise 50K news articles (512-dim) → L2-normalise → store in FAISS IndexFlatIP for cosine retrieval.
5	**Model Training**	LSTM (128→64 units) trained on 30-day sequences of 23 features; Huber loss; Adam optimiser; EarlyStopping; 80/10/10 split.
6	**Inference & RAG Query**	At prediction time: retrieve top-5 grounding news docs → supply as context → generate price forecast, volatility, and risk label.
7	**Evaluation**	Compute RMSE, MAE, MAPE, Directional Accuracy (%), Sharpe Ratio per stock.
8	**Visualisation**	Price trends, actual vs predicted, technical dashboards, sentiment–price correlation, risk dashboard, correlation heatmap.

6. RESEARCH GOAL & EXPECTED OUTCOMES

The primary research goal is to demonstrate that grounding LLM-based financial reasoning in a retrieved, verifiable document corpus (RAG) materially reduces hallucination and improves forecast reliability compared to unconstrained generative models. Specifically, the project aims to:

- Achieve directional accuracy ≥ 55% across all five stocks on the held-out test set.
- Produce risk-adjusted Sharpe Ratio > 1.0 for at least three equities.
- Show a statistically significant correlation ($|r| \geq 0.3$) between aggregated news sentiment and next-day returns.
- Demonstrate that RAG-retrieved context reduces generation of unsupported financial claims versus a baseline LLM with no retrieval.
- Deliver an explainable forecast engine — each prediction accompanied by retrieved evidence snippets, indicator values, and a natural-language rationale.

This mid-stage deliverable establishes a working data pipeline, feature store, sentiment layer, FAISS RAG index, and a preliminary LSTM model with measurable metrics. Subsequent phases will integrate a full FinBERT encoder, transformer-based time-series models (PatchTST / Informer), and an LLM-based explanation generator.